

ANTI-PROXY CLASSROOM ATTENDANCE SYSTEM

The Mini Project report submitted in partial fulfillment
of the requirements for the award of the Degree of
Bachelor of Technology

in

Computer Science and Engineering

under the

APJ Abdul Kalam Technological University

by

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “**Anti-Proxy Classroom Attendance System**” submitted by **Meera Krishna S (MGP23CS094)**, **Sulfa Saji (MGP23CS094)**, **Susan Saji (MGP23CS094)**, **Vasudev V (MGP23CS094)** for the award of the **Bachelor of Technology in Computer Science and Engineering** is a bonafide record of the Mini Project carried out by them under my guidance and supervision at “**Saintgits College of Engineering (Autonomous)**”.

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DECLARATION

We, Meera Krishna S (MGP23CS094), Sulfa Saji (MGP23CS094), Susan Saji (MGP23CS094), Vasudev V (MGP23CS094), hereby declare that this thesis entitled “**Anti-Proxy Classroom Attendance System**” is a record of the original work carried out by us under the guidance of **Er. Gayathri J L**, Department of Computer Science and Engineering, Saintgits College of Engineering. To the best of our knowledge, this work has not formed the basis for the award of any degree, diploma, fellowship, or any other similar title to any candidate in any University.

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Abbreviations

AI	–	Artificial Intelligence
ML	–	Machine Learning
YOLO	–	You Only Look Once
CNN	–	Convolutional Neural Network
MTCNN	–	Multi-task Cascaded Convolutional Neural Network
RGB	–	Red Green Blue
SQL	–	Structured Query Language
DB	–	Database

Abstract

Attendance monitoring is an essential component of academic evaluation in educational institutions. However, traditional methods such as manual roll calls and basic digital systems are time-consuming and vulnerable to proxy attendance, resulting in inaccurate records and reduced academic integrity.

The Anti-Proxy Classroom Attendance System is an AI-based solution designed to ensure accurate and secure attendance tracking by verifying the physical presence and identity of students. The system utilizes artificial intelligence techniques such as facial recognition and machine learning to authenticate students in real time, effectively preventing proxy attendance.

Attendance data is automatically processed and stored in a centralized database, enabling efficient monitoring and report generation while minimizing manual intervention. This proposed solution enhances transparency, accountability, and fairness in attendance management, providing educational institutions with a reliable, scalable, and technologydriven approach to classroom attendance

Chapter 1

Introduction

1.1 Background and Motivation

Attendance monitoring is an essential component in academic institutions as it directly influences student evaluation and discipline. Traditional attendance systems, such as manual roll calls and basic digital entry methods, are inefficient, time-consuming, and prone to manipulation.

One of the major issues in existing systems is proxy attendance, where students mark attendance on behalf of others. This compromises academic integrity and leads to inaccurate records.

With advancements in Artificial Intelligence, especially in computer vision and facial recognition, it has become possible to automate attendance systems with high accuracy. This motivated the development of the Anti-Proxy Attendance System, which uses AI-based face recognition to ensure authenticity and efficiency.

1.2 Problem Statement

Traditional attendance systems suffer from several limitations that affect both efficiency and reliability. One of the major issues is proxy attendance, where students mark attendance on behalf of others, compromising academic integrity. Manual methods such as roll calls are time-consuming and reduce valuable class time, while basic digital systems still depend heavily on human input. Additionally, these systems lack real-time verification mechanisms, making it difficult to ensure that the recorded attendance is accurate at the moment it is taken. As a result, attendance records often become inconsistent and unreliable.

Due to these challenges, there is a strong need for a secure, automated, and intelligent attendance system. Such a system should be capable of verifying student identity in real time, eliminating the possibility of proxy attendance, and reducing manual effort. By using advanced technologies like face recognition, the system can improve accuracy, ensure transparency, and provide a more efficient solution for attendance

1.3 Scope of the Project

The scope of this project focuses on developing an intelligent attendance system that uses real-time face detection and recognition to accurately identify students during class sessions. By automating the attendance marking process, the system significantly reduces the need for manual intervention, saving time and minimizing human errors. The use of facial recognition ensures that each student's identity is verified correctly, thereby effectively preventing proxy attendance and maintaining academic integrity.

In addition to attendance marking, the system includes a secure database for storing and managing attendance records in an organized manner. It also provides a user-friendly dashboard where both students and faculty can view attendance details, track records, and monitor performance. Although the system is mainly designed for classroom environments, its scalability allows it to be extended for use in larger institutions, organizations, or other domains where secure and automated attendance tracking is required.

Chapter 2

Literature Review

2.1 Introduction

The literature survey focuses on analyzing existing attendance management systems, including both traditional and AI-based approaches, to understand their features, strengths, and limitations. It helps identify gaps such as proxy attendance and lack of real-time accuracy, which provides a basis for designing an improved and more reliable automated attendance system.

2.2 Automated Attendance System Using Face Recognition (2019)

This system, proposed by Abhishek Kumar and Ankit Gupta, is based on Convolutional Neural Networks (CNN) for face recognition. The system captures images of students and processes them using CNN models to identify individuals and mark attendance automatically. The use of CNN helps in extracting facial features and improving recognition accuracy compared to traditional image processing methods.

One of the major advantages of this system is that it reduces the need for manual attendance marking and increases efficiency in classroom environments. However, the system faces significant challenges in real-time scenarios, especially in crowded classrooms. When multiple students are present in a single frame, the system struggles to accurately detect and recognize all faces. This leads to reduced performance and reliability. Additionally, the system may not handle variations in pose and background effectively, limiting its practical usage.

2.3 Smart Attendance System Using Deep Learning (2020)

This system uses advanced deep convolutional neural networks (Deep CNNs) to improve the accuracy of face recognition. It focuses on creating a smart and automated attendance system that can identify students based on trained datasets. The deep learning model learns complex facial features, making it more accurate than basic recognition systems.

The system performs well under controlled conditions, such as consistent lighting and stable camera positions. However, it is highly sensitive to environmental factors, particularly lighting variations. Changes in brightness, shadows, or low-light conditions can significantly affect the performance of the model. This makes it less reliable in real-world classroom environments where lighting conditions are not always stable. As a result, the system may fail to detect or correctly recognize faces in certain situations.

2.4 Face Recognition-Based Attendance System Using Deep Learning (2021)

This system introduces the concept of face embeddings combined with similarity matching techniques. Instead of directly comparing images, the system converts facial images into numerical feature vectors (embeddings) and compares them using similarity measures such as cosine similarity. This approach improves the accuracy of face recognition and reduces errors caused by minor variations in facial expressions.

Although the system provides better accuracy and robustness compared to earlier models, it has a major drawback in terms of computational complexity. Generating embeddings and performing similarity matching requires high processing power and memory. This makes the system computationally expensive and difficult to implement in real-time environments, especially on devices with limited hardware capabilities. As a result, its practical usability is limited in classroom scenarios where quick processing is required.

2.5 Real-Time Face Recognition-Based Attendance System (2021)

This system combines CNN and OpenCV to perform real-time face detection and recognition using live video streams. The integration of OpenCV allows the system to process video frames efficiently, making it suitable for real-time applications. The system captures video input, detects faces, and matches them with stored data to mark attendance instantly.

While the system is capable of real-time processing, it faces challenges when dealing with multiple faces in a single frame. In classroom environments, where many students are present, the system's accuracy decreases due to overlapping faces and variations in positioning. This results in missed detections or incorrect recognition. Additionally, the system may struggle with dynamic movements, such as students turning their heads or changing positions, which further affects accuracy.

2.6 YOLO-Based Real-Time Attendance Monitoring System (2022)

This system uses YOLO (You Only Look Once) for face detection along with deep learning models for recognition. YOLO is known for its high speed and efficiency in detecting objects in real time. By using YOLO, the system can detect multiple faces quickly and accurately, making it suitable for real-time classroom environments.

The major advantage of this system is its ability to handle multiple faces efficiently and perform fast detection. However, it requires a high-quality camera setup to achieve accurate results. If the camera resolution is low or the image quality is poor, the detection performance decreases significantly. Additionally, the system may require proper calibration and setup to work effectively, which can increase implementation complexity.

2.7 Summary

From the analysis of the above systems, it is clear that significant progress has been made in automating attendance using face recognition and deep learning techniques. These systems improve efficiency and reduce manual effort, but they still face several limitations.

Common issues include sensitivity to lighting conditions, high computational requirements, reduced accuracy in multi-face environments, and dependency on hardware quality. Many systems also lack strong mechanisms to completely eliminate proxy attendance.

The proposed Anti-Proxy Attendance System aims to overcome these limitations by combining YOLO for efficient face detection, FaceNet for accurate recognition, and multi-image verification to improve reliability. This approach ensures better accuracy, real-time performance, and effective prevention of proxy attendance.

Chapter 3

System Development

3.1 Objectives

The primary aim of this project is to develop an AI-based Anti-Proxy Attendance System that ensures secure, accurate, and efficient attendance management. The detailed objectives of the project are as follows:

1. **To analyze the limitations of existing attendance systems**

The project aims to study traditional attendance methods such as manual registers, RFID cards, and biometric systems. These systems often suffer from drawbacks like proxy attendance, time consumption, lack of real-time monitoring, and susceptibility to manipulation. Identifying these limitations helps in designing a more robust solution.

2. **To design an intelligent AI-based attendance system**

The objective is to create a smart attendance system that leverages Artificial Intelligence and Machine Learning techniques. The system will be designed to automate the attendance process, reduce human intervention, and provide a seamless and user-friendly experience.

3. **To implement face detection and face recognition techniques**

The project focuses on integrating computer vision algorithms to detect human faces and recognize them accurately. Face detection will identify the presence of a face in real-time, while face recognition will match it with stored data to verify identity.

4. **To prevent proxy attendance effectively**

A key objective is to eliminate the possibility of proxy attendance. By using real-time face recognition and verification, the system ensures that only the actual registered individual can mark attendance, thereby enhancing authenticity and reliability.

5. **To improve accuracy and efficiency of attendance management**

The system aims to minimize errors associated with manual entry and traditional methods. It ensures high accuracy in identification and faster processing time, making attendance recording more efficient and reliable.

6. To enable real-time monitoring and data management

The project also aims to provide real-time attendance tracking and maintain a centralized database. This allows easy access, monitoring, and management of attendance records by administrators.

7. To develop a scalable and secure system

Another objective is to ensure that the system can handle multiple users and large datasets while maintaining data security and privacy. Proper authentication and secure storage mechanisms will be implemented.

3.2 Proposed System

1 System Overview

The proposed system is an AI-based Anti-Proxy Attendance System designed to automate and secure the attendance process using advanced computer vision and machine learning techniques. The system integrates real-time face detection, face recognition, and database management to ensure accurate and reliable attendance recording. The system uses YOLO (You Only Look Once)-based face detection to identify faces in real-time through a camera feed. Once faces are detected, the system extracts facial features and converts them into face embeddings, which are numerical representations of facial characteristics. These embeddings are then compared with pre-stored embeddings in the database using similarity measures such as cosine similarity to identify the individual. To improve accuracy and reduce false positives, the system captures multiple face images (three per student) during each class period. These images are processed at different time intervals (e.g., start and end of the period), and attendance is finalized only if consistent detection is achieved. Based on this multi-image verification:

- A student is marked “Present” if detected consistently.
- Marked “Partial” if detected only once.
- Marked “Absent” if not detected.

The system is implemented using a backend developed with Flask, which handles authentication, attendance processing, and API services. A MySQL database is used to store student details, attendance records, and face embeddings. The system supports multiple user roles such as admin, faculty, and students, each with specific functionalities. Additionally, the system includes:

- Real-time live camera monitoring for administrators
- Offline attendance mode, where faculty can upload class images when the camera fails
- Automatic attendance marking per period, mapped with timetable data

- Camera health monitoring system to track failures and enable fallback mechanisms

Overall, the proposed system ensures a robust, scalable, and intelligent solution for modern attendance management.

3.3 Advantages Over Existing Anti-Proxy Attendance Systems

The proposed system offers several significant advantages compared to traditional and existing anti-proxy attendance systems:

1. **Complete Elimination of Proxy Attendance**

Unlike manual, RFID, or basic biometric systems, this system uses real-time face recognition, ensuring that only the actual student can mark attendance. This effectively prevents impersonation.

2. **Multi-Image Verification for Higher Accuracy**

The system captures multiple images during a class period and validates attendance based on repeated detection. This reduces errors caused by temporary occlusion, lighting issues, or misidentification.

3. **Fully Automated Attendance Process**

Attendance is recorded automatically without any manual intervention, saving time for both faculty and students and reducing administrative workload.

4. **Real-Time Processing and Monitoring**

The system provides live video streaming and real-time face detection, allowing administrators to monitor classroom activity and system performance instantly.

5. **Improved Accuracy Using AI Techniques**

By using face embeddings and similarity comparison methods, the system achieves higher accuracy compared to traditional biometric or card-based systems.

6. **Efficient Data Management**

Attendance data is stored automatically in a centralized database, enabling easy retrieval, report generation, and analysis.

7. **Robust Failure Handling Mechanism**

In case of camera or network failure, the system provides an offline mode where faculty can upload class images. This ensures continuity in attendance marking without data loss.

8. **Role-Based Access Control**

The system supports multiple user roles (admin, faculty, student), ensuring secure access and proper management of attendance data.

9. Scalability and Flexibility

The system can be extended to support multiple classrooms, institutions, and large numbers of students without significant performance degradation.

10. Transparency and Reliability

Since attendance is based on AI verification and stored digitally, the system ensures transparency and reduces disputes related to attendance records.

3.4 Design and Methodologies

1 System Design

The proposed Anti-Proxy Attendance System is designed as a comprehensive and intelligent solution that integrates modern web technologies with deep learning-based face recognition techniques to ensure accurate and secure attendance marking. The system follows a modular architecture consisting of a frontend interface, a backend server, a database, and machine learning models for face detection and recognition. The frontend is responsible for providing an interactive user interface for students, faculty, and administrators, enabling them to view attendance records, manage schedules, and monitor system activities. The backend is implemented using Flask, which handles all application logic, processes user requests, performs authentication, and coordinates communication between different modules of the system. The database layer is built using MySQL, which stores essential data such as student details, faculty information, attendance records, timetable schedules, and face embeddings. The system also incorporates advanced machine learning models, including YOLO for real-time face detection and FaceNet for generating unique facial feature representations. These components work together to ensure that attendance is marked accurately while preventing proxy attendance. The overall system architecture is designed to be scalable, efficient, and capable of handling real-time as well as offline attendance scenarios.

2 Methodology

The methodology for implementing the Anti-Proxy Attendance System follows a structured sequence from image acquisition to attendance recording. Initially, images are captured either through a live camera feed or uploaded classroom photographs, which serve as input to the system.

Face detection is performed using the YOLO algorithm, which efficiently detects human faces in a single pass by generating bounding boxes around relevant regions, eliminating unnecessary background information. The detected faces then undergo preprocessing, including cropping, conversion from BGR to RGB, resizing to 112×112 , and normalization. Face alignment may also be applied to correct orientation and improve recognition accuracy.

Feature extraction is carried out using the FaceNet model, which converts each face into

a numerical embedding representing its unique features. These embeddings are stored in a database during registration and used later for comparison.

During recognition, the system compares embeddings using cosine similarity. If the similarity score exceeds a predefined threshold, the person is identified and attendance is marked; otherwise, the face is treated as unknown, preventing proxy attendance.

Finally, attendance is recorded in the database with details such as student, date, and period. The system supports both real-time and offline attendance marking, and the recorded data is displayed on dashboards for students, faculty, and administrators. This approach ensures a reliable and efficient system by integrating computer vision, deep learning, and database technologies.

3 System Architecture

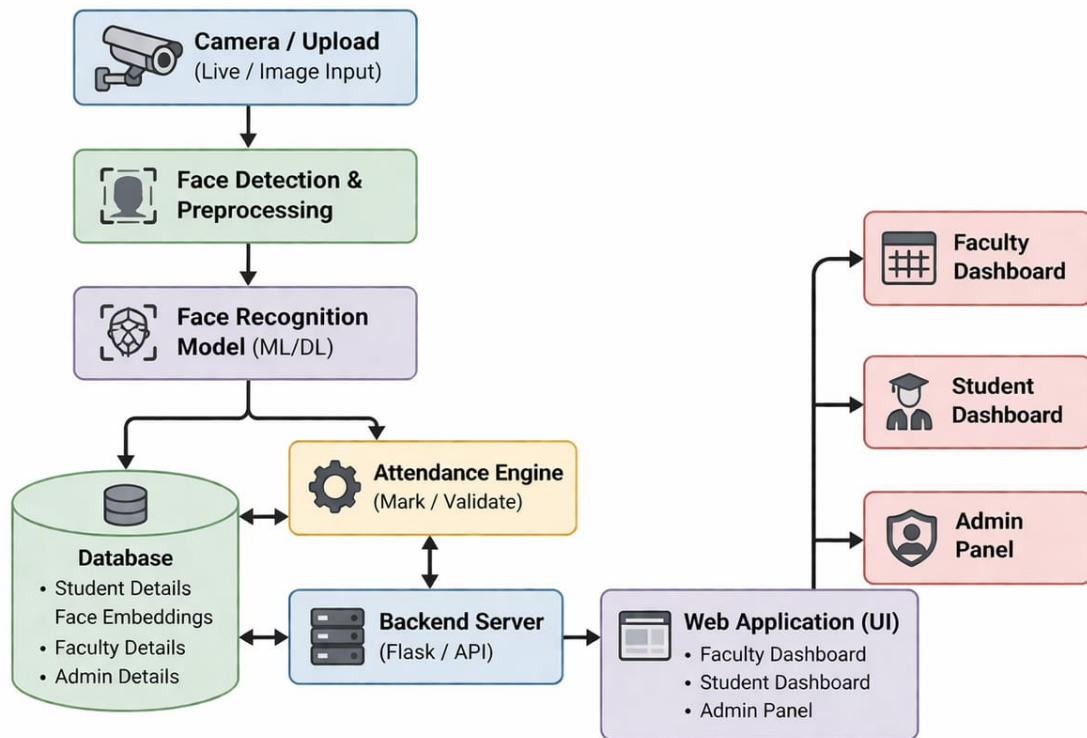


Figure B.3.1: System architecture

Chapter 4

Work Plan

This chapter presents the development strategy and execution plan of the proposed Anti-Proxy Attendance System. It includes the development model adopted, detailed phases of implementation, and visual representations through a Gantt chart and project timeline.

4.1 Development Model

The development of the proposed system follows an **Incremental and Iterative Model**. This approach allows the system to be built and refined in stages, enabling continuous testing and improvement at each phase.

Unlike traditional linear models, this methodology supports flexibility, making it easier to incorporate changes based on testing results and user feedback. Each module of the system—such as face detection, recognition, and web integration—is developed, tested, and validated independently before integration.

The key advantages of this model include:

- Early detection and correction of errors
- Continuous system improvement
- Better adaptability to changing requirements
- Parallel development of system components

4.2 Development Phases

The system development is divided into the following major phases:

1 Problem Analysis and Literature Review

This phase involved studying existing attendance systems and identifying their limitations, particularly issues related to proxy attendance and inefficiency. Various research papers and technologies such as face recognition and deep learning were analyzed.

2 Requirement Gathering and System Design

In this phase, system requirements were defined, including functional and non-functional requirements. The overall architecture of the system was designed, covering modules like face recognition, database management, and web interface.

3 Dataset Collection and Preprocessing

Facial images of users were collected under different conditions. These images were preprocessed to improve quality and consistency, including resizing, normalization, and alignment.

4 Face Detection and Recognition Development

This phase involved implementing the core functionality of the system. YOLOv8 was used for real-time face detection, while FaceNet was used to generate embeddings for recognition. Cosine similarity was applied for matching faces.

5 Frontend and Backend Integration

A web-based interface was developed for students, faculty, and administrators. The backend system was integrated with the face recognition module and database to enable real-time attendance marking and data management.

6 Testing, Evaluation and Documentation

The system was tested under different conditions to evaluate performance and accuracy. Errors were identified and corrected. Final documentation was prepared, including system design, implementation details, and results.

4.3 Gantt Chart

The Gantt chart shown in Figure B.0.1 illustrates the schedule and duration of each development phase.

4.4 Project Timeline

The timeline shown in Figure B.0.1 represents key milestones and their expected completion dates throughout the project lifecycle.

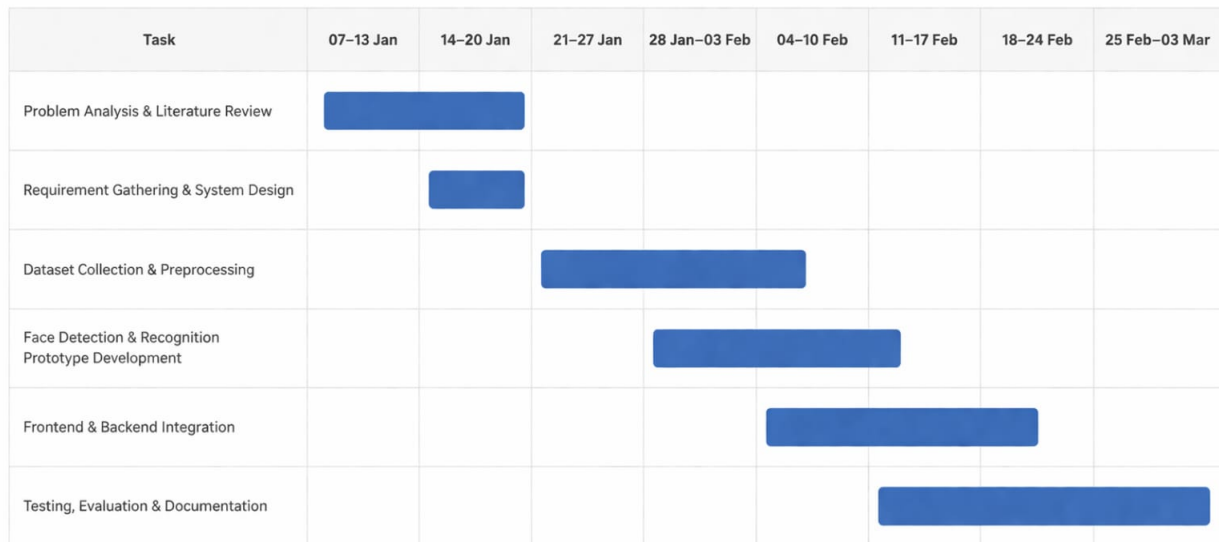


Figure B.0.1: Gantt Chart of Project Work Plan

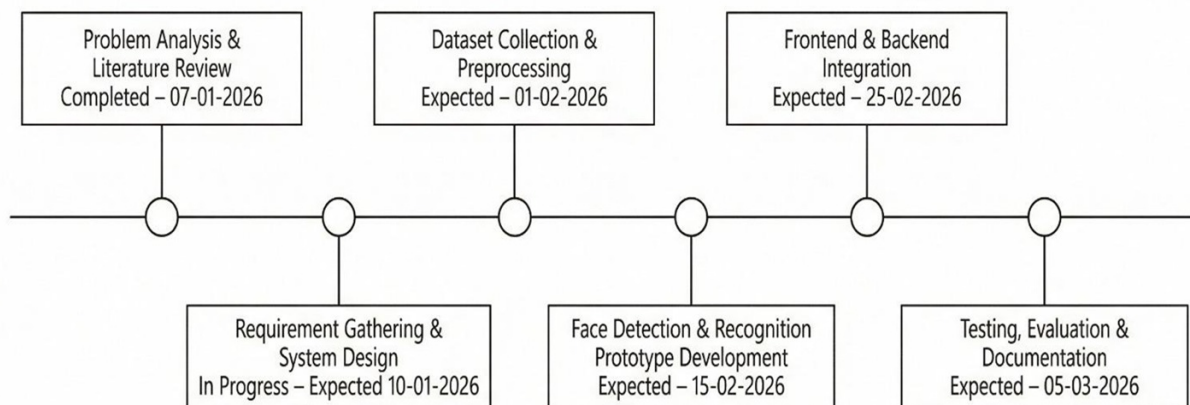


Figure B.0.1: Project Timeline and Milestones

Chapter 5

Result and Discussion

The proposed **Anti-Proxy Attendance System** was successfully developed and tested as an integrated solution combining face recognition, web technologies, and automated scheduling. The system demonstrated the ability to accurately detect and identify students using facial embeddings and mark attendance without manual intervention.

During testing, the system was deployed in a controlled classroom-like environment using a DroidCam-based camera setup. The face detection module based on YOLOv8 effectively identified multiple faces in real-time video streams, while the FaceNet model generated embeddings that enabled reliable identity matching using cosine similarity.

The automated attendance mechanism, which captures images at the start and end of each class period, functioned as intended. Students detected in both capture windows were marked as **Present**, those detected once as **Partial**, and those not detected were marked as **Absent**.

This dual-capture approach significantly reduced the possibility of proxy attendance.

The offline attendance feature also performed effectively under constrained conditions.

Faculty members were able to upload class images, and the system correctly processed them while enforcing business rules such as date restrictions and camera failure validation.

The web-based dashboards for students, faculty, and administrators provided clear visualization and control:

- Students could view attendance summaries and charts
- Faculty could manage attendance for assigned classes
- Admins could monitor system health and live camera feeds

Overall, the system achieved its primary objective of minimizing fraudulent attendance while reducing manual effort and improving transparency.

5.1 Challenges and Limitations

1 Technical Challenges

Face Recognition Accuracy in Real Conditions:

The performance of the face recognition system is sensitive to environmental factors such as lighting conditions, camera angle, and occlusions (e.g., masks, glasses). In low-light or crowded environments, detection accuracy may decrease, leading to missed or incorrect identifications.

Real-Time Processing Constraints:

Processing video frames in real time while performing face detection and embedding generation requires significant computational resources. Systems without GPU acceleration may experience latency, especially when multiple faces are present.

Camera Dependency and Network Stability:

The system relies on DroidCam for video input. Any change in IP address, network instability, or disconnection can interrupt the attendance process. Maintaining a stable connection between the camera and server is essential for reliable operation.

Embedding Threshold Sensitivity:

The cosine similarity threshold used for face matching must be carefully tuned. A lower threshold increases false positives (incorrect matches), while a higher threshold increases false negatives (missed detections). Achieving an optimal balance required iterative testing.

2 Potential Bottlenecks

Dataset Limitations:

The accuracy of the system depends heavily on the quality and diversity of stored face embeddings. Limited variations in facial expressions, angles, or lighting conditions can reduce recognition performance in real-world scenarios.

Scalability Issues:

As the number of students increases, the system must compare each detected face against a larger set of embeddings, which may lead to increased processing time and reduced performance.

Hardware Variability:

Differences in camera quality (resolution, frame rate) and system hardware can significantly affect detection and recognition accuracy, making performance inconsistent across setups.

Dependency on Continuous Background Execution:

The automated attendance system requires the background worker (`main.py`) to run continuously. Any interruption in this service may result in missed attendance sessions.

3 System Limitations

Security Concerns:

Currently, the system does not implement strong security practices such as password hashing or secure credential storage. This limits its readiness for production deployment.

Lack of Liveness Detection:

The system does not include anti-spoofing mechanisms (e.g., detecting photos or videos instead of real faces). This could potentially allow sophisticated proxy attempts.

Manual Intervention in Edge Cases:

In cases where the camera fails or recognition does not work properly, faculty intervention is required through the offline upload feature.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project successfully designed and implemented an AI-based Anti-Proxy Attendance System that leverages face recognition technology to automate and secure the attendance process. By integrating computer vision techniques with a web-based management system, the solution effectively addresses the issue of fraudulent attendance marking in educational institutions.

The system demonstrated the ability to:

Accurately detect and recognize student faces using YOLO-based detection and FaceNet embeddings
Automate attendance marking through scheduled capture sessions
Provide role-based access for students, faculty, and administrators
Support fallback mechanisms such as offline attendance upload under controlled conditions

The dual-phase attendance marking strategy (start and end of class) proved effective in reducing proxy attempts while ensuring fairness in attendance evaluation. Additionally, the system's dashboards and APIs improved usability, transparency, and administrative control. Overall, the project validates that AI-driven attendance systems can significantly enhance reliability, reduce manual workload, and improve institutional accountability when compared to traditional methods.

6.2 Future Scope

The current system provides a strong foundation, but several enhancements can further improve its performance, scalability, and security:

Advanced Liveness Detection: Integrating liveness detection techniques (e.g., blink detection, depth sensing, or anti-spoofing models) can prevent misuse through photographs or video-based proxy attempts.

Improved Recognition Accuracy: Expanding the training dataset with more diverse images (lighting, angles, expressions) and exploring advanced models such as ArcFace can improve recognition robustness.

Cloud Integration and Scalability: Migrating the system to a cloud-based architecture can enable better scalability, centralized data management, and support for multiple institutions simultaneously.

Mobile Application Integration: Developing a companion mobile application for students and faculty can enhance accessibility, allowing real-time attendance tracking and notifications.

Automated Model Updates: Implementing a mechanism to periodically update and retrain the face recognition model will help maintain accuracy as new students are added.

Enhanced Security Measures: Applying industry-standard security practices such as password hashing, encrypted database connections, and secure authentication mechanisms will make the system suitable for production deployment.

Multi-Camera and IoT Integration: Supporting multiple cameras across classrooms and integrating IoT-based smart devices can improve coverage and reliability in large institutions.

Real-Time Alerts and Analytics: Adding real-time alerts for absenteeism and advanced analytics (attendance trends, performance correlation) can provide deeper insights for administrators.

Appendix A

Stakeholder Information

A.1 Survey Analysis

1 Survey Overview

A stakeholder survey was conducted to understand real-world issues in classroom attendance systems. The survey focused on identifying challenges in existing attendance methods and gathering requirements for an improved solution.

Stakeholders included:

- Faculty members
- Academic staff

The survey helped identify key problems such as proxy attendance, manual effort, and lack of accuracy in traditional systems.

Survey Details:

- Survey Type: Online questionnaire
- Tool Used: Google Forms
- Number of Respondents: 6 faculty members
- Survey Duration: 7 days

2 Key Findings

The analysis of survey responses revealed the following observations:

- Majority of faculty reported issues with proxy and inaccurate attendance.
- Existing attendance methods are time-consuming and partially manual.
- Faculty showed strong interest in an AI-based automated attendance system.

- Most respondents believe the system will improve accuracy, fairness, and transparency.
- A secure, reliable, and automated system is strongly needed to eliminate proxy attendance.

A.2 Stakeholder Certification Letter

ANNEXURE-II

Stakeholder Validation Form

(Students print this and get it signed by the stakeholder)

Mini Project – B.Tech CSE
Department of Computer Science & Engineering
Saintgits College of Engineering

A. Student/Team Details

- **Project Title:** ANTI PROXY CLASSROOM ATTENDANCE SYSTEM
- **Team Members (with Roll Nos/ Register Nos):**
 1. Sulla Saju – 63
 2. Susan Saju – 64
 3. Vasudev V – 70
 4. Meera Krishna S – 75
- **Guide Name:** Er. Gayatri T.T

B. Stakeholder Details

- **Name of Stakeholder:** Aneesh.T
- **Designation/Role:** Assistant Prof., Mechanical.
- **Organization / Institution / Business:** Saintgits College of Engg.
- **Contact Information:**
 - **Phone:** 8904366760
 - **Email:** _____

o Address: _____

C. Nature of Involvement

(Stakeholder to mark ✓ where applicable)

- ☒ Problem context discussed and validated
- ☒ Requirements explained
- ☒ Feedback provided on preliminary prototype
- ☒ Final solution demonstrated and reviewed
- ☒ Stakeholder endorses the usefulness/relevance of the solution
- ☐ Other (Specify): _____

D. Stakeholder Comments (Mandatory)

(Stakeholder must write at least 3-4 lines on relevance, usefulness, accuracy, etc.)

As a teaching staff, several times I faced the issue of money. So, it will be better to develop a AI based anti-poody system.

E. Stakeholder Verification

I hereby verify that the student team has interacted with me regarding the above project, and the information provided is true to the best of my knowledge.

Signature: *Anesh*

Name: ANEESH . J

Date: 11.02.26

Official Seal / Business Stamp (if any): _____

A.3 Minutes of Meeting – Stakeholder Interactions

MINUTES OF MEETING – STAKEHOLDER INTERACTION

Mini Project – B.Tech CSE
Saintgits College of Engineering

- **Date of Meeting:** 11-2-2026
- **Location/Mode:** Saintgits College of Engineering (Autonomous) / offline
- **Stakeholder Name & Designation:** Dr. Sibhu K.R.
- **Project Title:** ANTI-PROXY CLASSROOM

ATTENDANCE SYSTEM

Discussion Summary
(Students should write 6–10 points.)

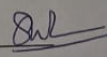
1. Need to prevent proxy attendance using face recognition
2. Use real-time face detection for marking attendance
3. System should support live camera and image input
4. Store student data and face embeddings in database.
5. Ensure accuracy, speed, and data Privacy.
6. can be integrated with college systems in future.

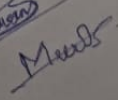
Decisions Taken

- Face Recognition will be used for attendance
- Database will store student details with embeddings.
- System will support live camera input

Action Items for Next Meeting

- Collect student face data for training
- Develop face detection & recognition module
- Design database and basic UI

Stakeholder Name & Signature: Dr. Shibu K.R 

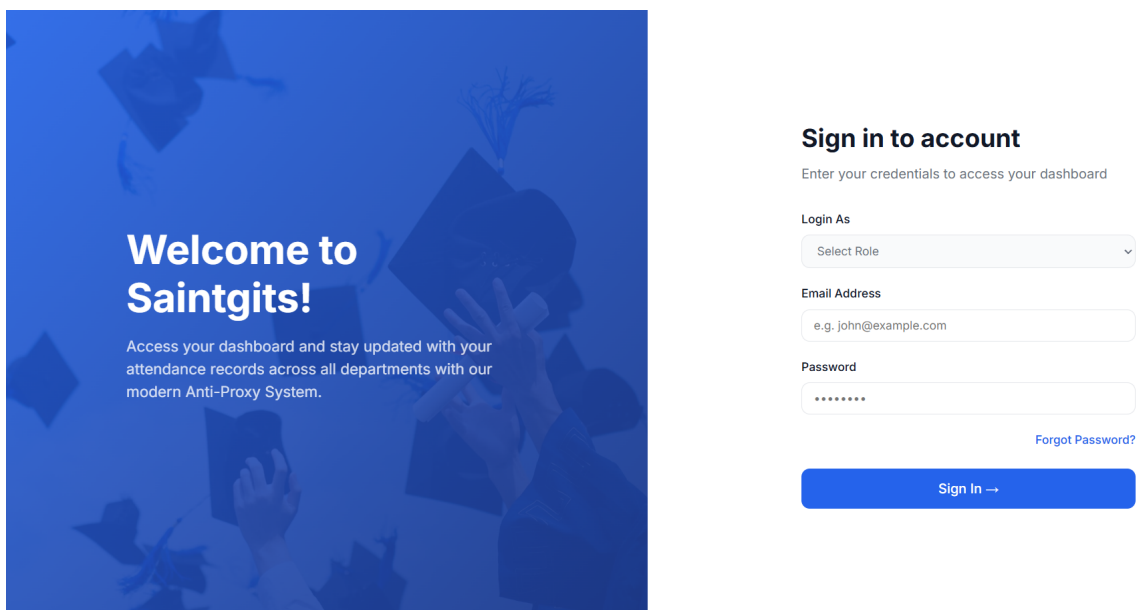
Student Names & Signatures: Vasudev V 
Sulfo Saji 
Susan Saji 
Meera Krishna S 

Appendix B

UI Screenshots

This appendix presents the user interface screens of the proposed Anti-Proxy Attendance System. The system includes separate modules for students, faculty, and administrators.

1 Login



The screenshot displays the login interface of the Anti-Proxy Attendance System. On the left, a blue-tinted image shows hands holding books, with the text 'Welcome to Saintgits!' and a description of the system's capabilities. On the right, a white login form is titled 'Sign in to account' and includes fields for 'Login As' (a dropdown menu), 'Email Address' (with a placeholder), and 'Password' (with masked characters). A 'Forgot Password?' link is located below the password field, and a blue 'Sign In →' button is at the bottom.

Welcome to Saintgits!

Access your dashboard and stay updated with your attendance records across all departments with our modern Anti-Proxy System.

Sign in to account

Enter your credentials to access your dashboard

Login As
Select Role ▼

Email Address
e.g. john@example.com

Password

[Forgot Password?](#)

Sign In →

Figure B.1.1: Login Interface

2 Student Module

Student Dashboard

Dashboard

My Timetable

Attendance History

Logout

Saintgits College of Engineering, Kottayam

Student Dashboard

Welcome back! View your academic information and attendance summary.

Student Details

Personal and academic information

Student Name

Joel Garvaziz

Roll Number

CSB-02

Registration Number

REGCSB-02

Batch

B

Semester

6

Email

cbs02@university.edu

Phone Number

None

Figure B.2.1: Student Dashboard

Student Dashboard

Dashboard

My Timetable

Attendance History

Logout

Saintgits College of Engineering, Kottayam

Attendance Overview Panel

Real-time summary from your recorded attendance

CL

Total Classes

12

PR

Classes Attended

3

AB

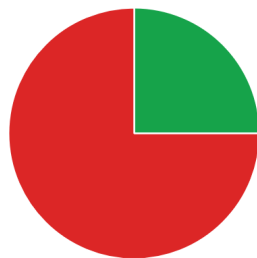
Classes Missed

9

AT

Attendance Percentage

25.0%



Present Absent

Present Percentage
25.0%

Absent Percentage
75.0%

Data Integrity
Chart values are generated from Present and Absent class counts.

Figure B.2.2: Student Timetable

Student Dashboard

Dashboard

My Timetable

Attendance History

Logout

Saintgits College of Engineering, Kottayam

DAY	PERIOD 1 9:00 – 10:00	PERIOD 2 10:05 – 11:05	PERIOD 3 11:15 – 12:15	PERIOD 4 13:15 – 14:15	PERIOD 5 14:20 – 15:20	PERIOD 6 15:30 – 16:30
MONDAY	Compiler Design	Algorithm Analysis and Design	Industrial Economics and Foreign Trade	Minor/Honors / Compiler Design (Remedial)	Compiler Design	Computer Graphics and Image Processing (Tutorial)
TUESDAY	Computer Graphics and Image Processing	Compiler Design (Tutorial)	Minor/Honors / Algorithm Analysis and Design (Remedial)	Algorithm Analysis and Design	Placement Training	Data Analytics / Programming in Python
WEDNESDAY	Compiler Design	Algorithm Analysis and Design	Industrial Economics and Foreign Trade	Networking Lab / Mini Project	Networking Lab / Mini Project	Networking Lab / Mini Project
THURSDAY	Data Analytics / Programming in Python	Computer Graphics and Image Processing	Data Analytics / Programming in Python (Tutorial)	Industrial Economics and Foreign Trade	Computer Graphics and Image Processing	Comprehensive Course Work
FRIDAY	Networking Lab / Mini Project	Networking Lab / Mini Project	Networking Lab / Mini Project	Free Hour	Algorithm Analysis and Design (Tutorial)	Minor/Honors / Advisory Hour

Figure B.2.3: Student Attendance Statistics

Student Dashboard

Dashboard

My Timetable

Attendance History

Logout

Saintgits College of Engineering, Kottayam

Attendance History

View your daily attendance record across all 6 periods.

DATE	PERIOD 1	PERIOD 2	PERIOD 3	PERIOD 4	PERIOD 5	PERIOD 6
2026-03-25	Present	Absent	Absent	Absent	Absent	Absent
2026-03-23	Present	Absent	Absent	Present	Absent	Absent

Figure B.2.4: Attendance History

3 Faculty Module

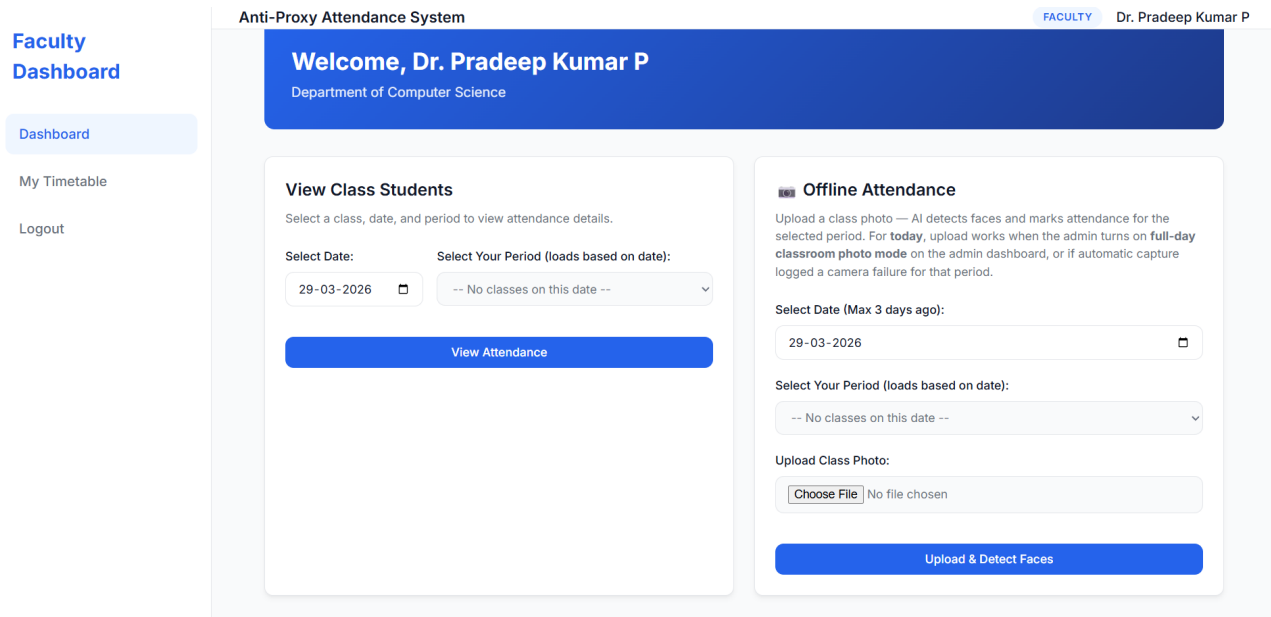


Figure B.3.1: Faculty Dashboard

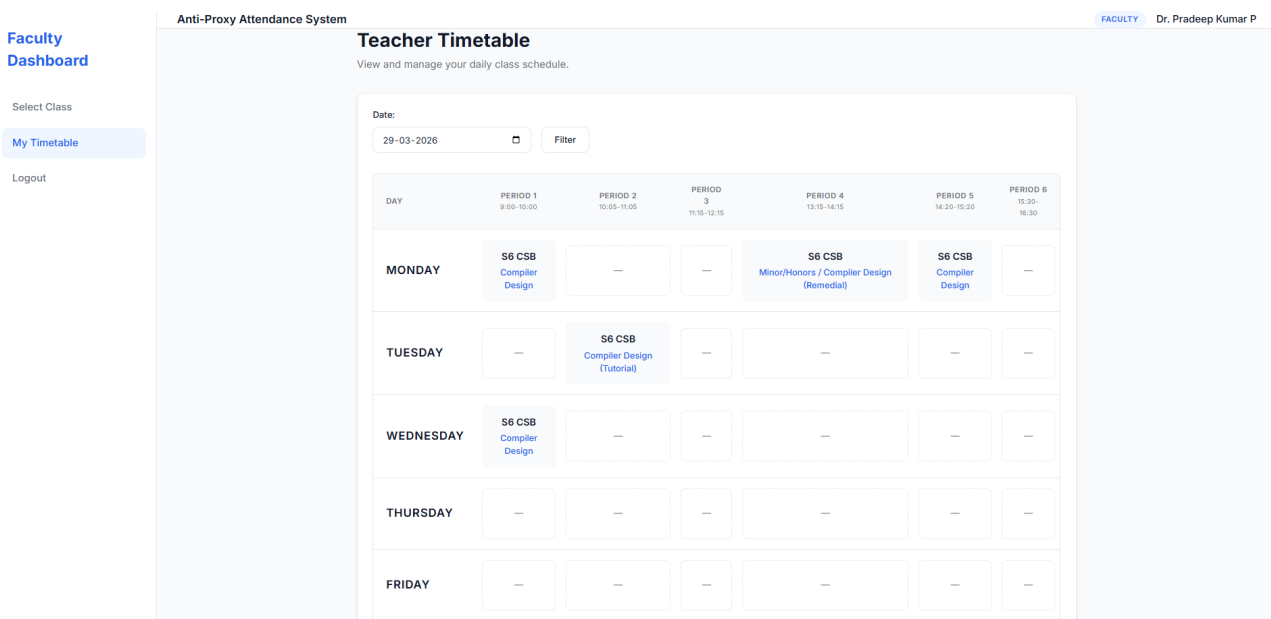


Figure B.3.2: Faculty Timetable

4 Admin Module

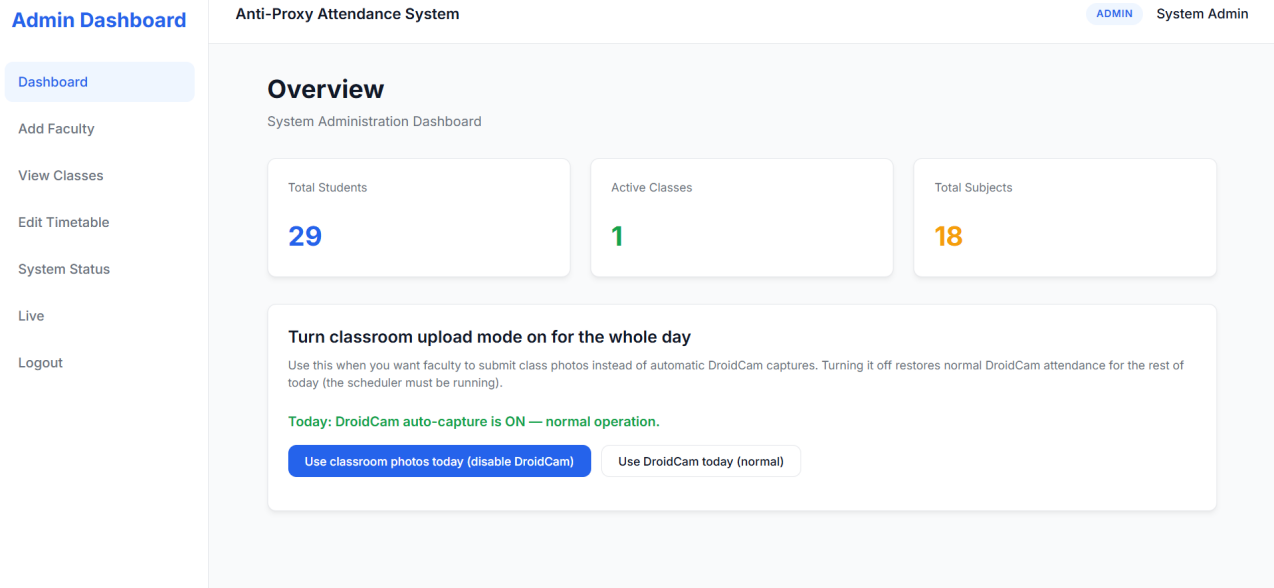


Figure B.4.1: Admin Dashboard

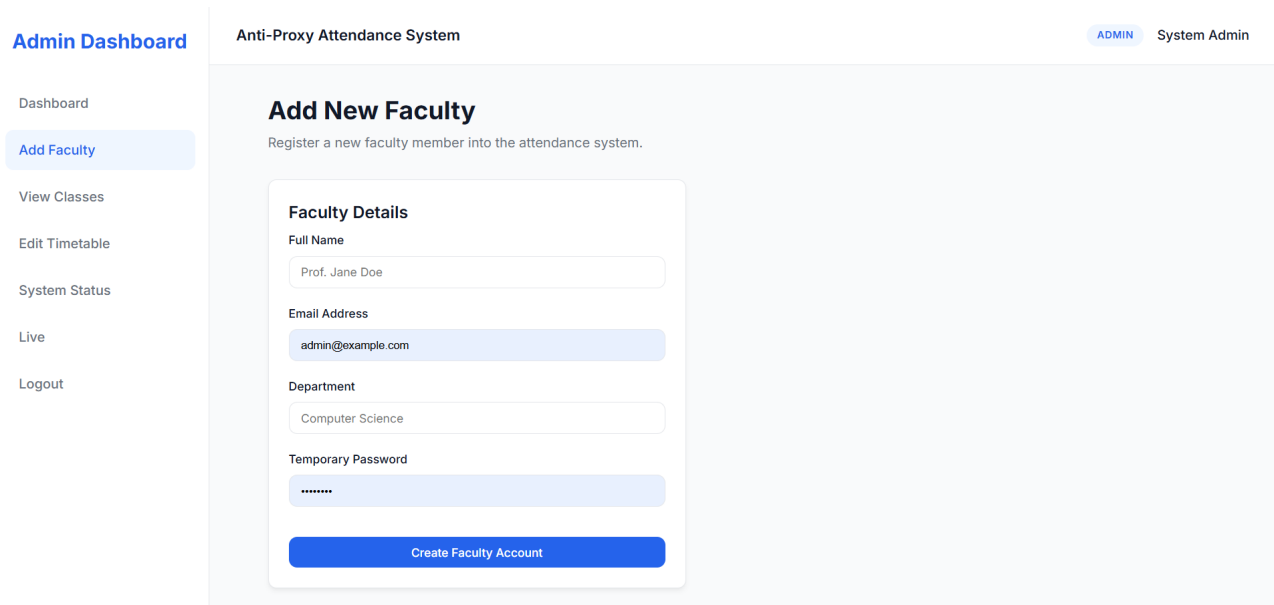


Figure B.4.2: Add New Faculty Interface

Admin Dashboard

- Dashboard
- Add Faculty
- View Classes
- Edit Timetable
- System Status
- Live
- Logout

Anti-Proxy Attendance System

ADMINSystem Admin

Class View

Select a class to view its students and timetable.

Select Class

Class:

S6 CSB

Students — S6 CSB

29 enrolled student(s)

#	NAME	ROLL NO	EMAIL
1	Jeswin Sabu	CSB-01	csb01@university.edu
2	Joel Garvaziz	CSB-02	csb02@university.edu
3	Joel Jose	CSB-03	csb03@university.edu
4	Lakshmi	CSB-04	csb04@university.edu

Figure B.4.3: Class View

Admin Dashboard

- Dashboard
- Add Faculty
- View Classes
- Edit Timetable
- System Status
- Live
- Logout

Anti-Proxy Attendance System

ADMINSystem Admin

Admin Timetable Editor

View and manage your daily class schedule.

Select Target Class:

S6 CSB

Edit Timetable

DAY	HOURL 1	HOURL 2	HOURL 3	HOURL 4	HOURL 5	HOURL 6
MONDAY	Compiler Design Suspend	Algorithm Analysis and Design Suspend	Industrial Economics and Foreign Trade Suspend	Minor/Honors / Compiler Design (Remedial) Suspend	Compiler Design Suspend	Computer Graphics and Image Processing (Tutorial) Suspend
TUESDAY	Computer Graphics and Image Processing Suspend	Compiler Design (Tutorial) Suspend	Minor/Honors / Algorithm Analysis and Design (Remedial) Suspend	Algorithm Analysis and Design Suspend	Placement Training Suspend	Data Analytics / Programming in Python Suspend
WEDNESDAY	Compiler Design	Algorithm Analysis and Design	Industrial Economics and Foreign Trade	Networking Lab / Mini Project	Networking Lab / Mini Project	Networking Lab / Mini Project

Figure B.4.4: System Health and Status

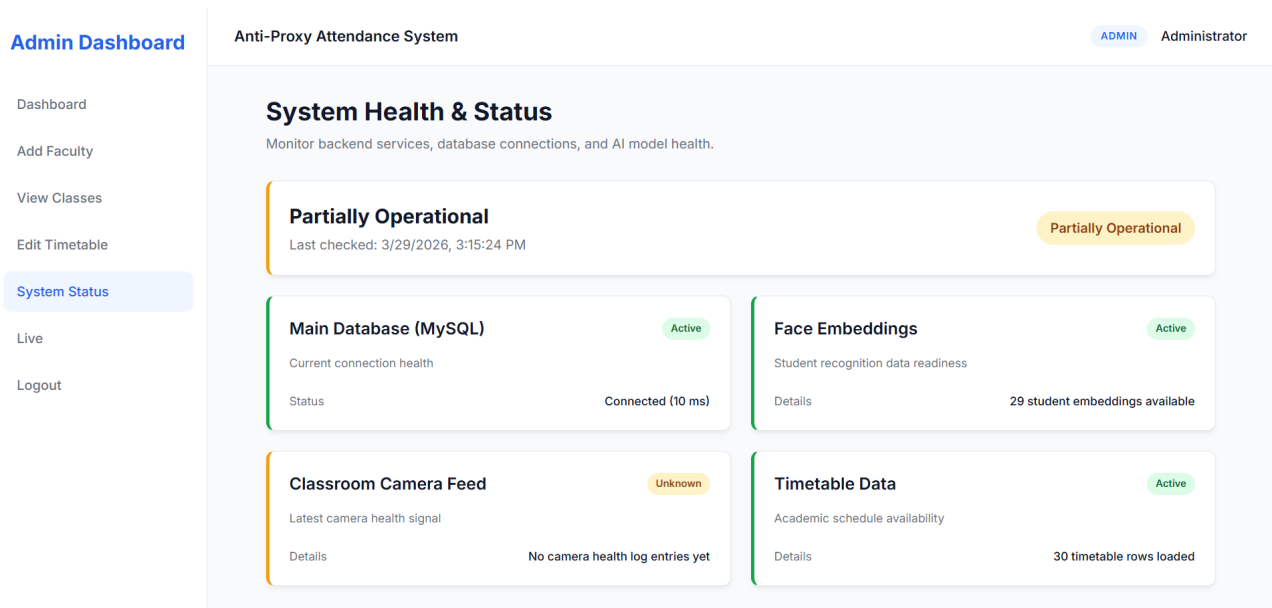


Figure B.4.5: Timetable Management

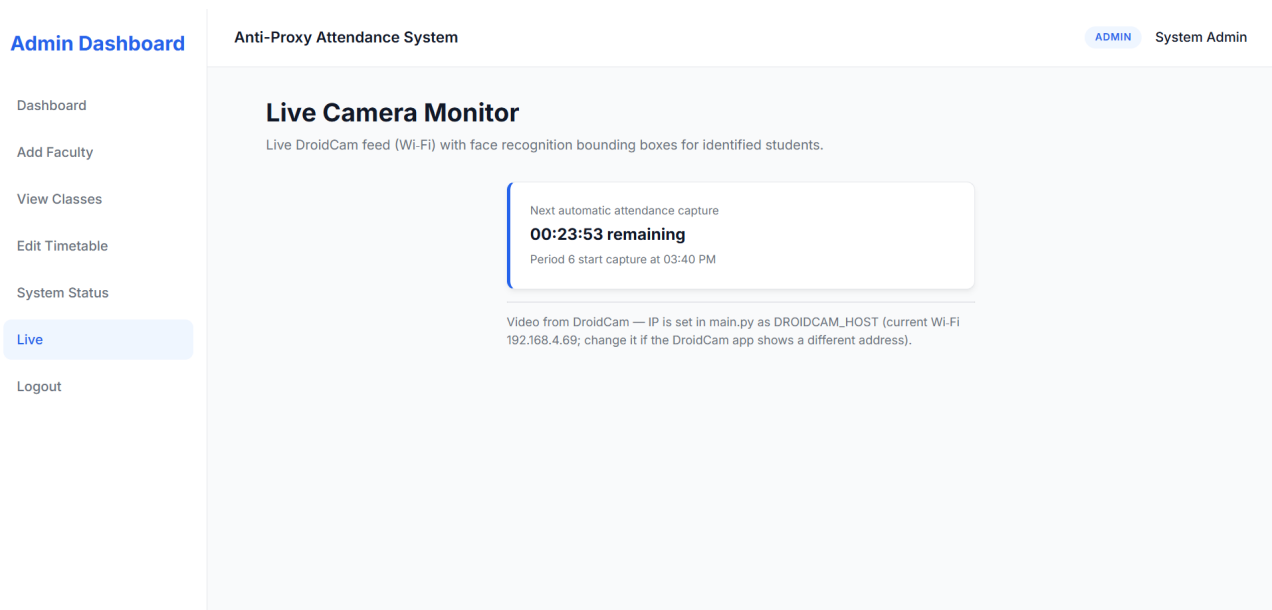


Figure B.4.6: Live Camera Monitoring

Appendix A

Source Codes

```
1 import os
2 import sys
3 sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
4
5 import cv2
6 import numpy as np
7 import mysql.connector
8 from models.embedding_model import get_face_embedding
9
10 DATASET_DIR = "datasets"
11
12 def get_db_connection():
13     return mysql.connector.connect(
14         host="localhost",
15         user="root",
16         password="",      #Mysql password Here
17         database="Attandance"
18     )
19
20 conn = get_db_connection()
21 cursor = conn.cursor()
22
23 print("Generating and saving embeddings to MySQL database...")
24
25 for class_name in os.listdir(DATASET_DIR):
26     class_path = os.path.join(DATASET_DIR, class_name)
27     if not os.path.isdir(class_path):
28         continue
29
30     print(f"Traversing Class Directory: {class_name}...")
31     roll_counter = 1
```

```

32
33 for student_folder in sorted(os.listdir(class_path)):
34     student_path = os.path.join(class_path, student_folder)
35     if not os.path.isdir(student_path):
36         continue
37
38     student_name = student_folder.strip().title()
39
40     class_prefix = class_name.split()[-1] if ' ' in class_name else class_name
41     roll_no = f"{class_prefix}-{roll_counter:02d}"
42
43     print(f" > Processing Student: {student_name} ({roll_no})")
44
45     person_embs = []
46
47     for img_name in os.listdir(student_path):
48         img_path = os.path.join(student_path, img_name)
49         img = cv2.imread(img_path)
50
51         if img is None:
52             continue
53
54         embedding = get_face_embedding(img)
55         if embedding is not None:
56             person_embs.append(embedding)
57
58         if len(person_embs) >= 3:
59             break
60
61     if person_embs:
62         master_emb = np.mean(person_embs, axis=0)
63         embedding_bytes = master_emb.astype(np.float32).tobytes()
64
65         student_email = f"{roll_no.lower().replace('-', '')}@university.edu"
66         default_pass = "cs123"
67
68         cursor.execute("SELECT student_id FROM students WHERE roll_no = %s",
69 (roll_no,))
70         result = cursor.fetchone()
71
72         if result:
73             cursor.execute(

```

```

73         "UPDATE students SET face_encoding = %s, name = %s,
target_class = %s, email = %s, password_hash = %s WHERE roll_no = %s",
74         (embedding_bytes, student_name, class_name, student_email,
default_pass, roll_no)
75     )
76     else:
77         cursor.execute(
78             "INSERT INTO students (roll_no, name, target_class, email,
password_hash, face_encoding) VALUES (%s, %s, %s, %s, %s, %s)",
79             (roll_no, student_name, class_name, student_email,
default_pass, embedding_bytes)
80         )
81
82         conn.commit()
83         roll_counter += 1
84
85 cursor.close()
86 conn.close()
87
88 print("Training complete - embeddings saved to database")

```

Appendix B

References

- [1] F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [3] Open Source Computer Vision Library, “OpenCV Documentation.” [Online]. Available: <https://opencv.org/>
- [4] Various authors, “IEEE Papers on Face Recognition Systems,” Research papers on biometric authentication and facial recognition technologies.
- [5] Anti Proxy Attendance System, “GitHub Repository.” [Online]. Available: <https://github.com/Vasu-uu/Anti-Proxy-Attendance-System>